Semi-Airborne Transient Electromagnetic Denoising Through Variation Diffusion Model

Fei Deng, Bin Wang, Peifan Jiang, *Graduate Student Member, IEEE*, Xuben Wang, and Ming Guo

***Abstract*— In geophysical exploration methods, the semi-airborne transient electromagnetic (SATEM) technique offers substantial exploration depth and the capability to acquire precise underground characteristics. However, the SATEM signal received by the induction coil has a low amplitude in the middle and late stages and is susceptible to various noise effects. Although a variety of conventional denoising methods are available today, their performance is often unsatisfactory, necessitating manual adjustments of the denoising outcomes. With the emergence of deep learning in various fields, several SATEM denoising neural network models have been proposed. However, these models are discriminative models and focus on modeling conditional probabilities, with insufficient generalization capabilities when faced with small datasets. On the other hand, the variational diffusion model (VDM) is a generative model that captures the joint distribution and exhibits excellent generalization capabilities. Building upon this premise, we explored a VDM-based denoising approach for SATEM. Nevertheless, due to the uncontrollable nature of the results generated by VDM, it is not possible to use VDM directly for denoising SATEM signals. To address this issue, we introduce the VDM-based denoiser with constraints, which incorporates guiding conditions to constrain the model and generate the desired denoising results, and propose a new SATEM signal denoising network called TEMSGnet. To further enhance the practical applicability of our method in real-world scenarios, we incorporate a wavelet transform-based supervised fine-tuning strategy. The experimental results demonstrate that TEMSGnet achieves effective denoising performance on both synthetic and real datasets. Furthermore, through inversion, TEMSGnet can more accurately approximate the true subsurface characteristics, thereby effectively reflecting the denoised outcomes. The code is available at https://github.com/woldier/TEMSGnet.**

***Index Terms*— Denoising, semi-airborne transient electromagnetic (SATEM), variational diffusion model (VDM).**

I. Introduction

THE semi-airborne transient electromagnetic (SATEM) method is a geophysical exploration technique that relies on the principle of electromagnetic induction. It derives from the transient electromagnetic (TEM) method. SATEM overcomes the limitations faced by TEM in conducting surveys in desert and desert-like regions by mounting the originally ground-based observation instruments onto an aircraft. Nevertheless, as with any method, the SATEM method has its strengths and weaknesses. While it effectively addresses the challenge of difficult ground observations, the signal strength received is often weak due to the observation equipment being positioned at a distance from the ground on the aircraft [1]. In addition, the further away the detection instruments are, the more susceptible the signal reception process becomes to disruption from environmental noise, artificial noise, and electronic device noise. Moreover, the amplitudes of early stage SATEM signal decay rapidly, while the mid-to-late-stage amplitudes decay slowly. Consequently, noise significantly affects the mid-to-late-stage signals, which correspond to the characteristics of the inverted intermediate-depth layers. Therefore, it is crucial to effectively remove noise from the SATEM signals so that subsequent inversion and other related geophysical research can be accurately conducted [2].

Numerous methods have been proposed for denoising SATEM signal, which can be broadly classified into two categories: traditional methods and deep learning-based methods. Traditional methods for SATEM signal denoising rely on wavelet transform and Kalman filtering, such as exponential adaptive window widthfitting algorithm (WEF) [3] and EF-AKF [4]. While these methods have attained satisfactory denoising outcomes to some extent, they come with certain limitations. First, achieving an optimal denoising effect necessitates the manual setting of multiple denoising parameters, which demands operators to possess extensive experience. Second, due to the small middle-to-late-stage amplitude of SATEM signals, traditional methods struggle to filter out the noise effectively, often requiring manual editing and adjustments. In addition, when dealing with a large volume of real SATEM data, the associated workload of manual adjustments becomes overwhelming.

Hence, considerable effort has been dedicated to utilizing deep learning techniques [5] to replicate the manual denoising process. In recent years, neural networks have been employed to address the challenge of SATEM denoising and have shown promising results. SFSDSAs [6] have been proposed as a solution for the SATEM denoising problem and have achieved certain denoising effects. Nonetheless, the fully connected architecture of the network poses a risk of overfitting to some extent, consequently hindering its generalization capability. To mitigate overfitting and address other concerns, a convolutional neural network (CNN)-based network called

Manuscript received 13 October 2023; revised 10 January 2024, 29 February 2024, and 9 April 2024; accepted 14 May 2024. Date of publication 17 May 2024; date of current version 31 May 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFB3905004 and in part by the National Science Foundation of China under Grant 41930112. *(Corresponding author: Bin Wang.)*

Fei Deng and Bin Wang are with the College of Computer Science and Cyber Security, Chengdu University of Technology, Chengdu 610059, China (e-mail: dengfei@cdut.edu.cn; woldier@foxmail.com).

Peifan Jiang, Xuben Wang, and Ming Guo are with the College of Geophysics, Chengdu University of Technology, Chengdu 610059, China (e-mail: jpeifan@qq.com; wxb@cdut.edu.cn; mingguo_cdut@163.com).

Digital Object Identifier 10.1109/TGRS.2024.3402212

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

TEMDnet [7] has been proposed. This network transforms the 1-D signal vector into a 2-D vector, enabling convolutional processing for signal denoising. TEMDnet effectively alleviates the overfitting phenomenon observed in the SFSDSAs network while enhancing the generalization ability of the network to a certain extent.

Convolutional networks, as referred to in TEMDnet, are discriminative models that capture the conditional distribution and learn the point-to-point mapping relationship within a given sample set. Thus, in order to achieve robust generalization, training on a large-scale sample is essential. However, due to the intricate nature of SATEM and the limited availability of training samples, obtaining densely and uniformly distributed training samples that cover the entire sample space is challenging. On the other hand, variational diffusion model (VDM) is a "likelihood-based" generative model [8] that models the joint distribution and demonstrates superior generative capability by predicting the mean and variance of the data [9]. It has been successfully applied in fields such as image and speech generation [10], [11]. Nevertheless, applying VDM directly to SATEM denoising problems is hindered by the uncontrollable nature of the generated results. To address this limitation, we conducted thorough research on VDM-based SATEM signal denoising. To further enhance the practical applicability of our method in real-world scenarios, we incorporate a wavelet transform-based supervised fine-tuning strategy. Specifically, for real-world data, we apply wavelet transform and manual adjustments to establish a small-scale authentic dataset representing real-world conditions. We then fine-tune the pretrained model on simulated datasets with this real-world dataset to obtain improved results. The contributions of this article are given as follows.

1) We proposed a constrained VDM, which extends the original VDM (origin-VDM) by incorporating constraint methods. By introducing constraints, we gain control over the generation process of the model, enabling us to input noisy signals and generate specific denoised signals through the constrained VDM. This facilitates the goal of denoising.
2) Building upon the constrained VDM approach, we proposed a novel SATEM signal denoising network known as TEMSGnet. This network not only exhibits excellent generalization capability but also demonstrates effective denoising performance.

II. Related Work

VDM is a development from autoencoders (AEs) [12] and variational AEs (VAEs) [13].

VAE does not directly encode the latent variable z through the encoder [14]. Instead, it learns the distribution of the hidden layer and utilizes the reparameterization tricks [13] to indirectly sample the hidden layer code z. The decoder then transforms z into the reconstructed output $\hat{x}$. The reparameterization tricks convert the encoding region of the latent space from discrete to continuous, expanding the coding region and reducing the distortion of the latent variable. In that case, the VAE model achieves better generative capability by possessing a larger and more expressive latent space. Fig. 1 depicts the fundamental structure of the VAE model. The encoder $q(z|x)$ defines the distribution of the latent space z given an observed input x, encompassing all potential observations. The decoder $p(x|z)$ defines the distribution from the latent space z to the real space x.

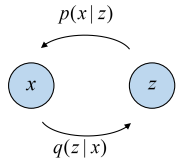


Fig. 1. Structure of VAE. The network structure diagram of VAE can be divided into two parts: encoder and decoder. The encoder $q(z|x)$ transforms the input x into the hidden variable z, while the decoder $p(x|z)$ transforms the hidden variable z into the output $\hat{x}$.

VAE, as a "likelihood-based" generative model, aims to generate $\hat{x}$ that closely resembles the distribution characteristics of the original data. Consequently, VAE seeks to maximize the log likelihood $\log p(x)$ of the generated $\hat{x}$. However, the direct maximization of $\log p(x)$ poses challenges [13]. To facilitate computation, the evidence lower bound (ELBO) [15] is commonly employed. By maximizing the ELBO, the objective of maximizing $\log p(x)$ can be achieved.

The hierarchical VAEs (HVAEs) [16] demonstrate enhanced flexibility in addressing real-world problems, owing to their capacity to learn deeper levels of latent space distributions. HVAE is structured hierarchically, comprising multiple VAEs stacked from top to bottom, and renowned for its high expressive power [17]. The hierarchical architecture of HVAE enables the capture of complex hierarchical representations, making it more proficient in representation learning compared to single-layer VAE [18]. If the encoding and decoding processes of each layer in HVAE solely rely on the previous layer, the overall encoding and decoding process form a Markov chain [19]. In such cases, it is referred to as a Markovian HVAE [20]. While HVAE offers high adaptability, optimizing both the encoder and decoder simultaneously becomes challenging due to the hierarchical nature of VAE layers. Thus, if we treat the decoder as a fixed-parameter process, effectively fixing the encoding process, the network only needs to optimize the parameters of the decoder. This approach reduces training difficulty.

If we model the decoder as a fixed-parameter process, effectively fixing the encoding process, the network can focus on optimizing the decoder parameters, thereby reducing training difficulty. To address this, the VDM was proposed as a solution. VDM can be considered a special case of a multilayer VAE [8], [9], [21]. In the VDM, the structure of the latent encoder at each time step is not learned; rather, it is predefined as a linear Gaussian model. In other words, it follows a Gaussian distribution centered around the output of the previous time step. The Gaussian parameters of the latent encoders vary over time in a manner that ensures the distribution of the latent variable at the final time step T becomes a standard Gaussian.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

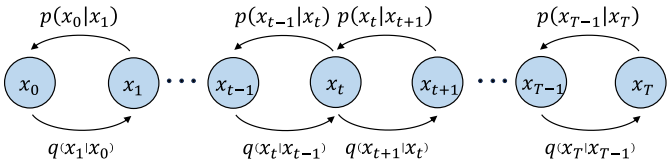


Fig. 2. Structure of VDM. The network structure diagram of VDM can be divided into multiple encoders and multiple decoders. The encoder $q(x_{t+1}|x_t)$ transforms the input x_t into the next hidden variable x_{t+1}, while the decoder $p(x_t|x_{t+1})$ transforms the hidden variable x_{t+1} into x_t.

Fig. 2 presents a visual representation of the VDM. In the diagram, x_0 represents actual data observations such as natural images, x_T represents the latent space variable at time T, and x_t is an intermediate latent space variable. Each transition $q(x_t|x_{t-1})$ of the encoder is modeled as a Gaussian distribution, with the mean being the output of the previous state. Mathematically, the encoder transitions are denoted by the following equation:

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)I\right). \tag{1}$$

The parameter α_t evolves over time according to a fixed or learnable schedule, designed such that the distribution of the final latent variable $p(x_T)$ follows a standard Gaussian. The joint distribution of the VDM decoder is represented by the following equation:

$$p(x_{0::T}) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$$
$$\text{where} \quad p(x_T) = \mathcal{N}(x_T; 0, I). \tag{2}$$

It is crucial to highlight that the encoder distribution $q(x_t|x_{t-1})$ does not possess any learnable parameters. Consequently, in the VDM framework, our attention is solely directed toward the conditional distribution $p_\theta(x_{t-1}|x_t)$. Analogous to VAE, the optimization of VDM involves maximizing the ELBO as depicted in the following equation [22]:

$$\begin{aligned}\log p(x) &\geq E_{q(x_{1:T}|x_0)}\left[\log \frac{p(x_{0:T})}{q(x_{1:T} \mid x_0)}\right] \\ &= \underbrace{E_{q(x_1|x_0)}\left[\log p_\theta(x_0 \mid x_1)\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}(q(x_T \mid x_0) \parallel p(x_T))}_{\text{prior matching term}} \\ &\quad - \underbrace{\sum_{t=2}^{T} D_{\mathrm{KL}}(q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t))}_{\text{denoising matching term}}.\end{aligned} \tag{3}$$

III. Method

A. Preliminary

In this study, we make the assumption that the complex environmental noise denoted as τ [7] ($\tau = b \cdot G(0, 1)$, where b is the noise level and $G(0, 1)$ indicates that it obeys a Gaussian distribution with mean 0 and variance 1; in the next formulation, we use the noise level expression abbreviated as $\tau = b$), where it acts as an additive component. The degradation process of the acquired noisy signal can be accurately represented by (4). In this equation, n represents the acquired signal with noise, while s represents the theoretical signal without noise. The main objective of our study is to recover the noise-free SATEM signal s from the SATEM signal n contaminated by noise

$$n = s + \tau. \tag{4}$$

B. VDM-Based Denoiser With Constraints

We propose a VDM-based denoiser with constraints to address the denoising problem in VDM. Our main objective is to guide the denoising process in the VDM framework by incorporating additional constraints. Our aim is to minimize a certain objective function and satisfy the given constraints to obtain the denoised signal. The introduction of these constraints into the VDM framework improves denoising performance and ensures the retention of desired characteristics in the denoised signal. These constraints can be derived from prior knowledge of the signal or specific properties that we aim to preserve during the denoising process. In particular, the noisy signal is used as a constraint in this article. Incorporating constraints into the VDM-based denoiser allows for effective noise removal while preserving crucial signal characteristics.

To enhance the denoising capability of the VDM-based denoiser with constraints, we propose the inclusion of an additional encoding process within the inference denoising phase. The purpose of this encoding process is to extract high-level representations from the noisy input signal. These representations capture meaningful information that can facilitate the denoising process and improve its effectiveness.

1) Encoder: It is common practice in the encoder to disregard the variances of the forward process and consider them as fixed constants through reparameterization [13]. In our implementation, the approximate posterior distribution q does not involve any learnable parameters [9]. Hence, it can be treated as a constant and ignored during the training process. As the encoder is Markovian at each time step, the posterior distribution of the encoder can be expressed as follows:

$$q(x_{1:T}|x_0) = \prod_{t-1}^{T} q(x_t|x_{t-1}) \tag{5}$$

where x_0 represents the observed data in the ground truth space and x_t , $t \in [1, T]$, represents the corresponding latent space variable.

The encoding process defines that each latent space representation x_t follows a Gaussian distribution with its previous latent variable x_{t-1}. To parameterize the Gaussian encoding process, we define the mean $\mu_t(x_t) = (\alpha_t)^{1/2}x_{t-1}$ and the variance $\Sigma_t(x_t) = (1-\alpha_t)$, where α_t is a time step-dependent hyperparameter. Mathematically, the encoder transformation can be expressed as (1). By employing the reparameterization trick, we can encode x_0 into the latent variable x_t for any time step $t \in [1, T]$ using (6), assuming that x_0 is known. In this equation, α_t can vary with increasing time steps based on a fixed temporal schedule, leading to the final latent variable following a standard Gaussian distribution denoted as $x_T \sim \mathcal{N}(x_t; 0, I)$:

$$\begin{aligned}x_t &= f_{\text{encoder}}(x_0, t, \varepsilon) \\ &= \sqrt{\bar{\alpha}_t}x_0 + \left(1-\bar{\alpha}_t\right)\varepsilon \quad ; \quad \text{where } \varepsilon \sim \mathcal{N}(\varepsilon; 0, I).\end{aligned} \tag{6}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

2) Decoder: In (3), we analyzed the VDM optimization function, which models the data distribution $p(x)$. However, for the SATEM denoising problem, our objective is to model the conditional data distribution $p(x|y)$, where y represents the constrained information. We aim to accurately generate the desired data based on the given conditional information y. To achieve this, a straightforward approach is to include the conditional information alongside the current time step information as inputs to the decoder at each layer. Consequently, there are changes to the input of the network, requiring a reconsideration of the model to accommodate the conditional information. Taking into account the incorporation of the constraint information, (2) can be rewritten as follows:

$$p(x_{0:T}|y) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t, y)$$
$$\text{where} \quad p(x_T) = \mathcal{N}(x_T; 0, I). \tag{7}$$

The ELBO optimization objective, formulated to maximize (3), can be reformulated as (8). The first term in the equation represents the reconstruction term, which, similar to the single-layer VAE, can be approximated and optimized using Monte Carlo methods. The second term, known as the prior matching term, measures the similarity between the encoder of the network distribution and the true latent space distribution (standard Gaussian). Since the encoding process produces a standard Gaussian distribution, this term can be assumed to be zero. The third term, referred to as the encoder–decoder matching term, aims to align the decoding process $p_\theta(x_{t-1}|x_t, y)$ with the encoding and subsequent decoding of the actual signal $q(x_{t-1}|x_t,x_0)$. Minimizing their Kullback–Leibler (KL) divergence allows the decoding steps to closely resemble each other [22]

$$\begin{aligned}\log p(x|y) &= \log \int p(x_{0:T}|y)dx_{1:T} \\ &\geq \underbrace{\mathbb{E}_{q_{(x_1|x_0)}}\left[\log p_\theta(x_0|x_1, y)\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}\left[q(x_T|x_0) \parallel p(x_T)\right]}_{\text{prior matching term}} \\ &\quad - \underbrace{\sum_{t=2}^{T} \mathbb{E}_{q_{(x_t|x_0)}}\left[D_{\mathrm{KL}}\left(q\left(x_{t-1}|x_t,x_0\right) \parallel p_\theta(x_{t-1}|x_t, y)\right)\right]}_{\text{decoder matching term}}. \end{aligned} \tag{8}$$

According to Bayes' theorem, the reparameterization trick, and (6), we can derive the distribution as follows, denoted by (9). $q(x_{t-1}|x_t, x_0)$ follows a Gaussian distribution, where its mean $\mu_q(x_t, t)$ is a function of x_t and t and its variance $\Sigma_q(t)$ is a function of α_t. We can rewrite $\Sigma_q(t)$ as (10)

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}\left(x_{t-1}; \underbrace{\frac{1}{\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon_0}_{\mu_q(x_t,t)}, \underbrace{\frac{(1-\alpha_t)\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}I}_{\Sigma_q(t)}\right) \tag{9}$$

$$\Sigma_q(t) = \sigma_q^2(t) = \frac{(1-\alpha_t)\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}. \tag{10}$$

In order to match $p_\theta(x_{t-1} \mid x_t, y)$ with $q(x_{t-1} \mid x_t, x_0)$, we also model $p_\theta(x_{t-1} \mid x_t, y)$ as a Gaussian distribution. Therefore, we can construct the mean of $p_\theta(x_{t-1} \mid x_t, y)$ to be similar to $q(x_{t-1} \mid x_t, x_0)$ while ignoring the variance, which can also be learned. We set the variance of $p_\theta(x_{t-1} \mid x_t, y)$ to be consistent as $\Sigma_q(t) = \sigma_q^2(t)$. The distribution followed by $p_\theta(x_{t-1} \mid x_t, y)$ can be expressed in (11), as shown at the bottom of the next page. The Gaussian distribution of $p_\theta(x_{t-1} \mid x_t, y)$ in this equation is similar to that in VDM, but the predicted input of the network needs to incorporate additional constraint information y. The conditional information can be any type, and in this article, it refers to the noisy signal n.

Therefore, maximizing the ELBO of the constrained VDM is equivalent to minimizing the third term, which is the decoder matching term of the ELBO, as shown in the following equation:

$$\begin{aligned}&\arg\min_\theta D_{\mathrm{KL}}\left(q\left(x_t|x_{t-1,}x_0\right) \parallel p_\theta(x_{t-1}|x_t, y)\right) \\ &= \arg\min_\theta \frac{1}{2\sigma_q^2(t)}\frac{(1-\alpha_t)^2}{\left(1-\bar{\alpha}_t\right)\alpha_t}\left\|\epsilon_0 - \hat{\epsilon}_\theta(x_t, t, y)\right\|_2^2. \end{aligned} \tag{12}$$

Thus, we can optimize a model by predicting $\hat{\epsilon}_\theta(x_t, t, y) \approx \epsilon_0$ into (11) to obtain its mean and variance. This can be further used to obtain x_{t-1} through reparameterization, as shown in the following equation:

$$\begin{aligned}x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\hat{\epsilon}_\theta(x_t, t, y) \\ + \frac{(1-\alpha_t)\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}\varepsilon. \end{aligned} \tag{13}$$

3) Proposed New SATEM Signal Denoising Network—TEMSGnet: The implementation of the backbone network in the constrained VDM does not have specific requirements. Previous studies have predominantly employed Unet as the backbone network [8], [9]. However, since signals are 1-D and Unet is designed for 2-D image inputs, signals cannot be directly used as input for Unet. Converting the signals into 2-D images may lead to the loss of structural and continuity information. Alternatively, employing a new 1-D–2-D method can be complex [7]. Hence, to overcome these challenges and achieve improved performance, this article introduces a novel network architecture named TEMSGnet, which is specifically designed for signal processing.

The architecture of the network is depicted in Fig. 3. Fig. 3(a) presents an overview of the network architecture. The network predicts $\hat{\epsilon}_\theta(x_t, t, y)$ and calculates $\hat{x}_{t-1}$ using (13) given the input x_t, t, and y. The "In Conv" operation applies a 1D-convolution with a kernel size of 3, a padding of 1, and one input channel to the 64-length input signal. This operation increases the dimension from 1 to 32 channels. The "Embed" layer encodes the time step t into a representation of size 1×64. The "Out Conv" operation transforms the output of the "UpBlock" into a single-channel representation, which serves as the final output. The green circles with a "+" sign represent the concatenation of channels. The operation labeled as "1D-Conv" in Fig. 3(b) corresponds to a 1D-convolution with a kernel size of 3 and a padding of 1. The input and output channels remain unchanged. The "Down Sample" operation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

serves as a downsampling operation that doubles the number of channels and reduces the input length by half. The pink circles with a "+" sign denote the addition of values, symbolizing the presence of a residual connection [23]. In Fig. 3(c), the only difference between the details of the "Up Block" and the "Down Block" is the use of the "Up Sample" operation in the last layer. The purpose of the "Up Sample" operation is to reduce the number of channels in the input vector by half while doubling its length.

The modules shared by the down block and up block are as follows.

1) *Group Normalization:* Group normalization (GN) [24] is an alternative to traditional batch normalization (BN) [25] that addresses the issue of poor performance with small batch sizes. When using small batch sizes, BN may not effectively estimate the mean and variance of the entire data distribution. In addition, BN may yield suboptimal results for tasks with highly imbalanced binary classification. To overcome these limitations, GN divides the channels into groups and performs normalization within each group. Instead of computing the mean and variance based on the batch size, GN computes the mean over $((C/G)) \times H \times W$, where C is the number of channels, G is the number of groups, H is the height, and W is the width. By doing so, GN decouples the normalization from the batch size and allows the network to achieve better performance. By normalizing within groups, GN provides a more robust estimation of statistics and helps mitigate the negative impact of small batch sizes or imbalanced data distribution.
2) *Scale Shift:* In order to fuse the features generated by the 1D-convolution with the time step t, this study performs an embedding operation on t to obtain its embedding result. The features generated by the 1D-convolution are then subject to a scale shift operation, which incorporates the information of the time step t into the features. The specific steps of the scale shift operation are illustrated in Fig. 4. In the scale shift operation, the input of the time embedding undergoes a linear projection to match the length of the input. Then, the number of channels is increased to match the dimensionality of the input, and a copy is made. One copy is scalar multiplied with the input, and the result is added to the other copy. This process allows the time step information to be fused with the features through scaling and shifting operations, enabling the model to incorporate temporal information into the feature representation.
3) *Linear Attention:* The use of linear attention mechanism in this study aims to capture the temporal dependencies and establish connections between consecutive signals in the time domain. Unlike RNN networks [26], which may encounter limitations in capturing long-term dependencies due to the influence of sequence length, the linear attention mechanism allows for a more direct and efficient extraction of global and local contextual information [27], [28]. Compared to CNN and RNN architectures, the linear attention mechanism offers several advantages. First, it requires fewer parameters, which reduces the overall complexity of the model. Second, it enables the model to effectively capture temporal relationships and dependencies across the entire signal, without being limited by sequence length. This makes it particularly suitable for processing time-domain signals where capturing temporal correlations is crucial. By incorporating the linear attention mechanism, the model can enhance its ability to attend to relevant temporal information and improve the overall performance in capturing the underlying patterns and dependencies within the time-domain signals.
4) *Residual Learning:* The use of residual learning is an effective technique to further enhance feature learning in deep CNNs. Deeper CNN architectures have the capability to capture more abstract features and improve performance in tasks such as classification and regression. However, simply increasing the depth of a network does not always lead to better performance, as it can suffer from issues such as vanishing or exploding gradients [23]. Residual learning, introduced in the context of the ResNet architecture, addresses these problems by incorporating residual blocks that allow for the direct propagation of information from earlier layers to later layers. By using skip connections, residual learning enables the network to learn residual mappings, which can help mitigate the degradation problem associated with increasing network depth. These residual connections provide shortcuts for gradient flow, making it easier for the network to optimize the deeper layers and improve overall performance. In addition to addressing gradient-related issues, residual learning has been observed to enhance the network performance. The skip connections allow for the preservation of important information from earlier layers, enabling the network to effectively learn complex representations. This approach has demonstrated improved convergence and achieved state-of-the-art results in various deep learning tasks.

Fig. 5 illustrates the forward process and inverse process of VDM with constraints. In the forward process of a VDM, data evolve from its initial state x_0 to a target state x_T. This progression occurs through a series of controlled transformations

$$p_\theta(x_{t-1}|x_t, y) = \mathcal{N}\left(x_{t-1}; \underbrace{\frac{1}{\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\hat{\epsilon}_\theta(x_t, t, y)}_{\mu_\theta(x_t,t,y)}, \underbrace{\frac{(1-\alpha_t)\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}I}_{\Sigma_q(t)}\right) \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

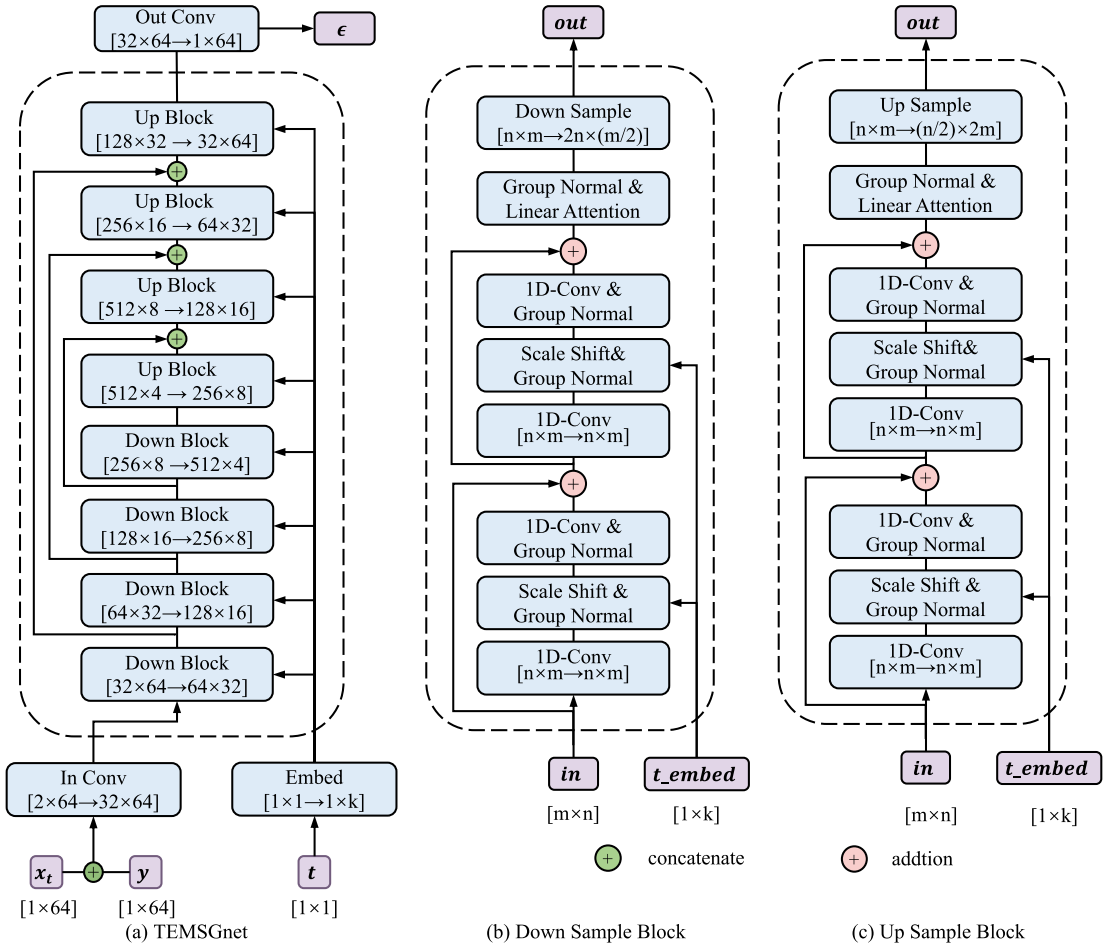


Fig. 3. Architecture of the TEMSGnet. (a) Overview of the network architecture that has four down blocks and four up blocks. (b) Overview of the down block architecture. (c) Overview of the up block architecture. The green circles with a "+" sign represent the concatenation of channels. The pink circles with a "+" sign denote the addition of values, symbolizing the presence of a residual connection [23].

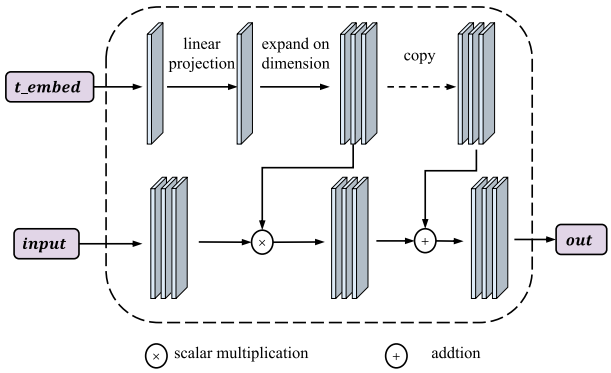


Fig. 4. Architecture of scale shift. The scale shift operation involves linearly projecting and duplicating the time step embedding, followed by scalar multiplication and addition. This process effectively fuses time step information with 1D-convolution-generated features, enabling the model to incorporate temporal details into the feature representation.

without altering the fundamental nature of the data. It is akin to guiding data through a sequence of structured adjustments, leading to a modified representation at x_T. In the reverse process, starting from x_T and returning to x_0, the data revert through a sequence of reverse transformations. The goal is to recover the original state x_0 while adhering to constraints, notably constraint y. The constraint, when applied to the reverse process, ensures that the recovered data retain specific characteristics or features represented by y. This process effectively retraces the steps of the forward transformation but in a controlled manner to maintain adherence to the imposed. (Note that during training, x_0 represents s, and during testing, it represents n.)

The training and denoising inference specific process for the denoising algorithm in the network is given in Algorithms 1 and 2, respectively.

Algorithm 1 Training

repeat

$x_0 = s \sim q(x_0)$; $y = n$

$t \sim Uniform(\{1, \ldots, T\})$

$\varepsilon \sim \mathbf{N}(0, I)$

$x_t = f_{encoder}(x_0, t, \varepsilon)$

take gradient descent [21] step on

$$\frac{1}{2\sigma_q^2(t)} \frac{(1-\alpha_t)^2}{(1-\bar{\alpha}_t)\alpha_t} \| \epsilon_0 - \hat{\epsilon_\theta}(x_t, t, y) \|_2^2$$

until converged

IV. Experiment and Analysis

A. Dataset

To overcome the time-consuming process of creating training samples from collected SATEM data and the need for a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

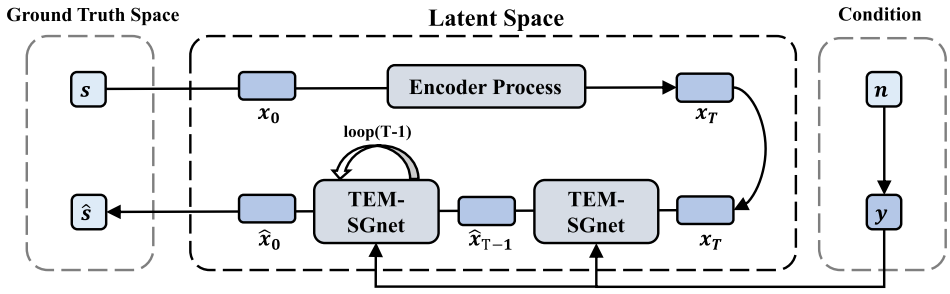


Fig. 5. Forward process and inverse process of VDM with constraints. In the forward process, data evolve from x_0 to x_T through controlled transformations. In the reverse process, the goal is to return from x_T to x_0 while adhering to constraint y [corresponding to n in (4)]. During training, x_0 represents s, and during testing, it represents n. This controlled reverse transformation maintains specific characteristics represented by y, retracing steps to recover the original state.

Algorithm 2 Denoising

$x_T = f_{encoder}(x_0, T, \varepsilon)$; $y = n$
for $t = T, \ldots, 1$ **do**
  $z \sim N(0, I)$ **if** $t > 0$, **else** $z = 0$
  $x_{t-1} = \frac{1}{\sqrt{\alpha_t}} x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}} \hat{\epsilon}_\theta(x_t, t, y) + \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \varepsilon$
end for
return x_0

large number of samples to train an effective pretrained denoising model, we have employed forward modeling with noise to generate a substantial simulated dataset for pretraining. Nevertheless, a disparity between real and simulated data remains, posing a challenge to effectively apply the pretrained model for denoising real data. To tackle this issue, we incorporate manual editing of a small portion of sideline data while processing real data, forming a transfer learning dataset used for transfer learning. This transfer learning process enables the model to effectively denoise the majority of the remaining data.

1) Synthetic Dataset: To enhance the similarity between the pretrained model and real data, we designed 16 fundamental geological models that capture common resistivity distribution characteristics. These models comprise two-layer, three-layer (with high- or low-velocity anomalies), four-layer (with one or two high- or low-velocity anomalies), and five-layer (with multiple high- or low-velocity anomalies) configurations. These models have a standardized depth of 300 m. Fig. 6(a) illustrates the designed geological models [1].

During the subsequent forward modeling process, our simulations relied on these geological models, and we introduced random variations to the depth values range $(-50, 50)$ and resistivity values (range $(-100, 100)$ for models with overall low resistivity and -500 to 500 for models with overall high resistivity) within a predefined range for each layer. The randomized depth values are added to each layer's original depth values in the base model. The depth values of the base model's layers are all set to be greater than 50 meters, ensuring that the additional values drawn from the range $(-50, 50)$ are also greater than zero. For instance, consider the first model in Fig. 6(a), which is a two-layer model. The depth of the first layer is approximately 90 m. Assuming a random layer depth of -50 m, the adjusted depth of the first layer becomes 40 m, still greater than zero and not extending above the surface.

In addition, we defined the forward modeling parameters based on commonly used values in semi-airborne electromagnetic field surveys. These parameters were selected to vary randomly within a reasonable range, as presented in Table I. Using randomization, a total of 32 000 forward modeling results were generated to adhere to realistic geological conditions, incorporating varying geoelectric models and forward modeling parameters. Fig. 6(b) illustrates the noise-free SATEM curves associated with the geological models. Following that, training sample pairs were acquired by introducing noise to the SATEM curves within the time domain.

In this study, we introduced Gaussian noise with varying levels of intensity. We selected Gaussian noise due to its prevalence in real exploration scenarios, where it typically impacts the signal. For signals with low noise levels ($\tau = 10$ nT), it can be observed that the theoretical curve in the time domain is almost indistinguishable from the noisy curve. The SATEM signal curves in both the time and logarithmic domains, along with their corresponding noise curves, are shown in Fig. 7. However, in the logarithmic domain, the secondary field component is more affected by noise interference. Therefore, the evaluation of denoising performance in the subsequent methods is based on the logarithmic domain.

We employ the simulated SATEM signals generated in Section IV-A1 as our dataset. Within this dataset, we designate the SATEM signals produced by models 10 and 14 in Fig. 6(a) for use in the validation set, while the SATEM signals generated by the remaining models constitute the training set. Subsequently, to align with the network's requirements, we introduce noise at the appropriate levels to the noise-free signals, as defined by (4), yielding the necessary pairs of samples for training and validation across various noise levels. It is important to emphasize that the added noise in our study adheres to a Gaussian distribution.

2) Real Dataset: Owing to the intricate nature of real data, using a network model trained solely on simulated data for real data denoising may not yield optimal results, despite constructing a large and diverse pretraining dataset through geological forward modeling. Thus, we conducted traditional denoising methods and manual fine-tuning on a small subset of real data to obtain accurately denoised real data. Subsequently, we employed transfer learning by training the pretrained network using the real dataset. Afterward, the transfer learning model was utilized for denoising the real data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

TABLE I

FORWARD MODELING PARAMETERS. NOTE THAT PARAMETERS ARE UNIFORMLY DISTRIBUTED WITHIN THE GIVEN BIAS

	Coil area	Sideline heigh	Current intensity	Horizontal distance of measurement point from the center of the line source	Vertical distance of measurement point from the center of the line source	Time range	Number of time windows
default	10 (m^2)	20 (m)	20 (A)	$(-1500m, 1500m)$	200 (m)	$5.20833e^{-6} - 0.005(ms)$	64
bias	± 5 (m^2)	± 10 (m^2)	± 10 (A)	± 1000 (m)	± 50 (m)	–	–

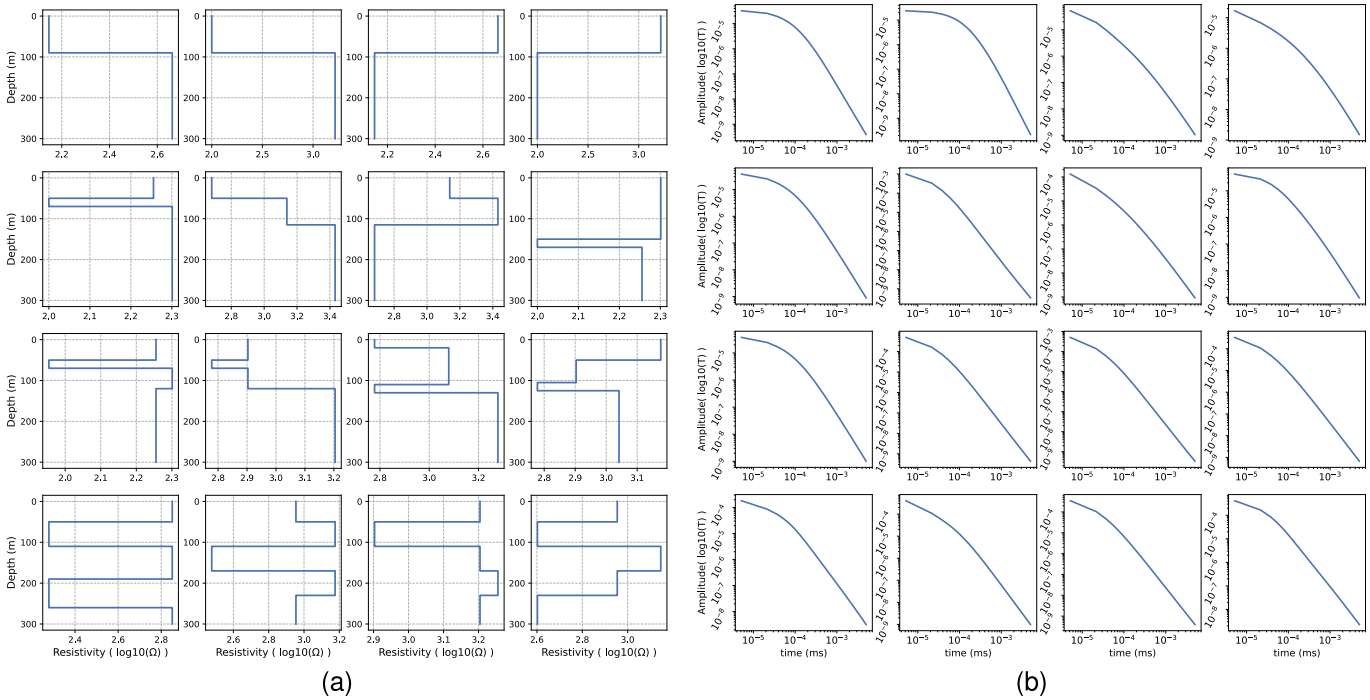


Fig. 6. (a) SATEM geological model diagram with different anomalies. (b) Forward modeling of SATEM signal curves.

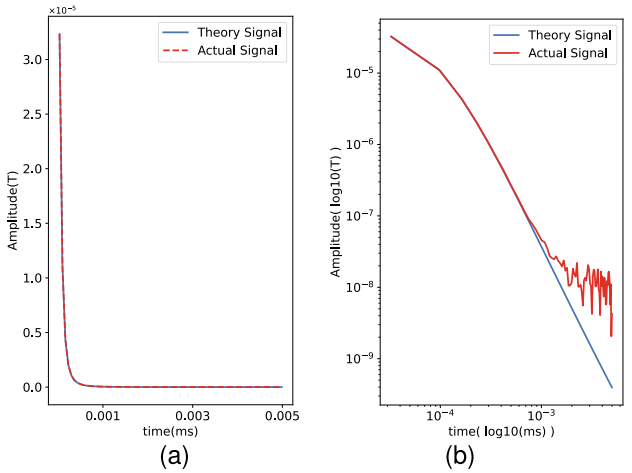


Fig. 7. Theoretical SATEM signal and noisy SATEM signal in (a) time domain and (b) logarithmic domain.

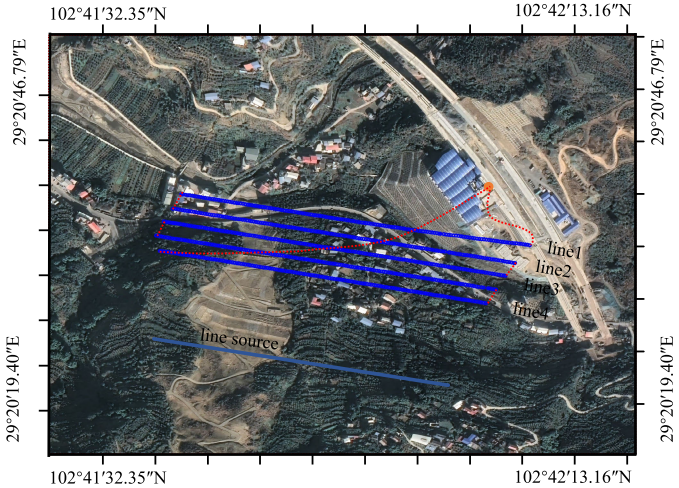


Fig. 8. Actual region for the SATEM signal acquisition. The actual effective collection point accounts for 600 m, which is the width of the area where the blue dotted line is located. The distance between the survey lines (blue dotted line) is 50 m. The length of each survey line is 800 m. The blue solid line represents the position of the line source, and it should be noted that the length of the line source is 800 m. In addition, Orange dots represent the takeoff position of the vehicle. The vehicle flies in a clockwise direction.

The specific location of the exploration area is shown in Fig. 8, located at 29°20′19.40″E–29°20′46.79″E and 102°41′32.35″N–102°42′13.16″N. The center of the source line is located at (29°20′34.9″E, 102°41′47.5″E), the direction is parallel to the blue line, and the length is 600 m. The height of the source line is 20 m and the current intensity is 20 A. The transmitter model is TXU30. The specification of receiver and z-axis inductive magnetic field sensor is shown in Table II.

B. Training Details

The network proposed in this article was implemented using PyTorch. We set the time steps to 1000 and optimized the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SPECIFICATIONS OF RECEIVER AND z-AXIS INDUCTIVE MAGNETIC FIELD SENSOR

Receiver Specifications	
Total Channels	2
Gain Options	0.25, 1, 4, 16
Sampling Rate	32kHz
Maximum Input Signal Range	-2.5V to 2.5V
ADC Bit Depth	24-bit
Dynamic Range	>120dB
Synchronization Method	GPS and Temperature-Compensated Crystal Oscillator
Attitude Detection Accuracy	Three-axis ±0.05°
Positioning Accuracy	Post-differential processing, centimeter-level
Controller and Processor	ARM and DSP
Specifications of the Z-axis Inductive Magnetic Field Sensor	
Effective Area	20000 m^2
Resonance Frequency	9.7kHz
Frequency Noise Level	3 pT/$\sqrt{}$Hz@1Hz 0.1 pT/$\sqrt{}$Hz@10Hz 1fT/$\sqrt{}$Hz@1Hz
Time Noise Level	1mV@20kHz (18 nT/s) 83uV@1kHz (1.4 nT/s)

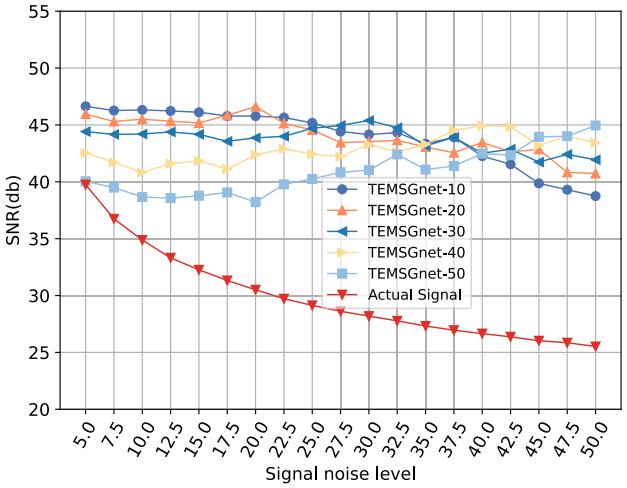


Fig. 9. SNR results obtained from models (different colored curves) trained on different datasets (10, 20, 30, 40, and 50 nT) and tested on different valid datasets ranging from 5 to 50 nT (at intervals of 2.5). The points where the curves are different in the horizontal direction represent testing on valid datasets with different noise levels. Since we can superimpose the signal with the noise by (4), it is possible to superimpose the datasets in the training set to obtain noisy signals with different noise levels for testing.

entire model for 2000 epochs using the Adam optimizer [29]. Set the batch size to 64. The training was performed utilizing an NVIDIA GeForce RTX 3080 graphics processing unit (GPU).

C. Baseline Methods and Quantitative Metric

In this study, we undertook a comprehensive evaluation of various denoising methodologies that have exhibited efficacy in the specialized domain of SATEM signal denoising using well-established benchmarks. Specifically, our comparative analysis involved benchmarking against established methods, namely, exponential fitting-adaptive Kalman filter (EK-AKF) [4], SFSDSAs [6], and TEMDnet [7].

To broaden the scope of our investigation and provide a holistic perspective on generative models, we extended our comparative analysis to include C-GAN [30], a notable generative network. This addition enables a more nuanced understanding of the proposed method's performance concerning other state-of-the-art generative networks.

Signal-to-noise ratio (SNR) was employed as a quantitative evaluation metric to compare the denoising performance of different methods. Equation (14) illustrates the calculation of SNR

$$\mathrm{SNR} = 10 \times \log_{10} \frac{\|s\|_2^2}{\left\|\hat{n} - s\right\|_2^2} = 10 \times \log_{10} \frac{\|s\|_2^2}{\left\|\hat{\tau}\right\|_2^2} \tag{14}$$

where s denotes the original noise-free signal and $\hat{n}$ represents the denoised signal predicted by the network. A higher SNR value signifies superior denoising performance.

D. Results

1) Denoising on Noise Levels of a Large Range on Simulated Dataset: The TEMSGnet demonstrates excellent adaptability in processing input signals with varying levels of noise. In particular, it can generate accurate predictions for signals with a diverse range of noise levels using a well-trained network model. Fig. 9 illustrates the SNR for the denoising results achieved by the trained models on different noise level valid datasets. In order to harness the complete potential of the simulated dataset and its relevance to noise signals of varying levels, multiple models were trained across a range of noise levels ($\tau = \{10, 20, 30, 40,$ and 50 nT$\}$). Then, for each model trained at different noise levels, it was tested on various valid sets with different noise levels ranging from 5 to 50 nT (with an interval of 2.5).

In Fig. 9, the different colored curves show the performance comparison of the models trained at different noise levels. Each curve in the horizontal direction represents the denoising performance of the model on the validation set with different noise intensities. It visually represents the efficacy of the models in reducing noise and enhancing signal quality. The derived conclusions from these observations are outlined as follows.

1) The models trained at specific signal levels are not only effective for the current noise level but also for a wide range of noise levels in the signals. This phenomenon can be explained by the fact that TEMSGnet learns the distribution, allowing it to have good generalization ability.
2) The models exhibit strong performance in terms of SNR not only when the valid dataset noise level closely matches that of the training set noise level the model uses but also across other varying noise levels datasets. This robust performance can be attributed to the incorporation of conditional information y within the network architecture, enabling the model to reconstruct signals based on this additional data. Upon closer examination, we conclude that the observed consistency across different noise levels is indicative of our model's resilience in effectively managing variations in noise levels.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

TABLE III

PERFORMANCE OF DIFFERENT METHODS ON SIMULATED SATEM SIGNALS AT NOISE LEVELS $\tau = 5$ NT, $\tau = 10$ NT, AND $\tau = 20$ NT. THE VALUES REPRESENT THE SNR IN DECIBELS (DB), AND THE BEST DENOISING RESULT IS HIGHLIGHTED IN BOLD. NOTE THAT ALL THE DEEP LEARNING METHODS ARE TRAINING UNDER A SINGLE NOISE LEVEL OF 10 NT

Method	Training	$\tau = 5(nT)$	$\tau = 10(nT)$	$\tau = 20(nT)$
EK-AKF	Needless	44.94	38.22	33.75
SFSDSA	Need	45.64	40.41	38.67
C-GAN	Need	45.94	40.87	39.25
TEMDnet	Need	46.37	42.91	40.11
TEMSGnet	Need	**46.67**	**46.32**	**45.77**

3) Based on the above two points, it can be concluded that this network has strong generalization ability and can reconstruct the clean signal based on the conditional information (i.e., the noisy curve), thus enabling effective denoising. The model demonstrates good performance on both the training and testing sets, making it applicable for denoising real SATEM signals.

2) Comparison With Other TEM Signal Denoising Method on Simulated Dataset: To compare with other methods (EK-AKF, SFSDSA, C-GAN, and TEMDnet) for SATEM denoising, all deep learning methods were trained on a simulated dataset with a Gaussian noise level of $\tau = 10$ nT. Subsequently, we assess the denoising performance using simulated datasets with varying noise levels ($\tau = \{5, 10,$ and 20 nT$\}$), mimicking the network's ability to denoise real-world data.

In Table III, the observed results are as follows.

1) All methods show good SNR levels at $\tau = 5$ nT, which is because the noise level is low, resulting in less noticeable denoising effect. All deep learning methods (SPSDSA, TEMDnet, C-GAN, and TEMSG-net) perform well at $\tau = 10$ nT, which matches the noise level in our training dataset.
2) At noise levels $\tau = 5$ nT, both C-GAN and TEMSGnet show competitive results, as the noise level is low, resulting in less noticeable differences in denoising effects. As the noise level increases to $\tau = 10$ nT, TEMSGnet outperforms C-GAN, showcasing its ability to effectively denoise signals under conditions similar to the training dataset. This trend continues at higher noise levels ($\tau = 20$ nT), where TEMSGnet consistently demonstrates superior denoising capabilities compared to C-GAN.
3) TEMSGnet outperforms SFSDSA, C-GAN, and TEMDnet at noise levels $\tau = 5$ nT, $\tau = 10$ nT, and $\tau = 20$ nT. Conversely, the performance of the SFSDSA, C-GAN, and TEMDnet is not as good as TEMSGnet at these noise levels. This is due to the diverse nature of real-world noise in SATEM signals. The modeling of data distribution in TEMSGnet enables better generalization of the network.
4) Overall, the proposed method in this article demonstrates better robust denoising capabilities compared to existing deep learning-based methods in complex real-world exploration scenarios.

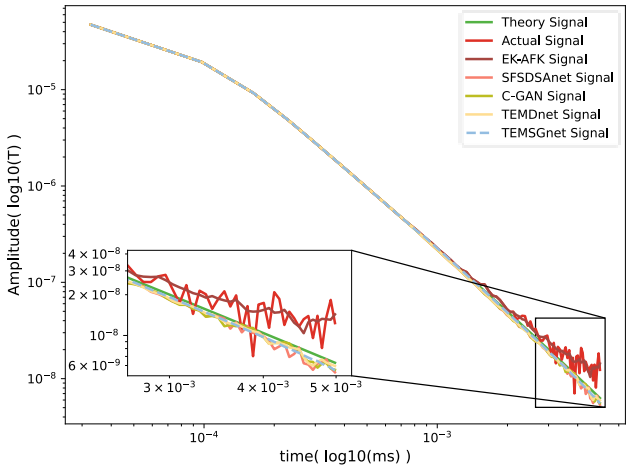


Fig. 10. Comparison of the theoretical noiseless signal and the actual signal and methods' denoising results in the logarithmic domain. The actual signal represents the noisy signal with a noise level of 10 nT, the theory signal represents the theoretical noiseless signal, and the rest of the curves represent the denoised signals of different networks trained on the training set with a noise level of 10 nT.

In order to further illustrate the denoising performance of all methods, visual examples of specific denoising results are presented in Fig. 10. Fig. 10 displays the simulated SATEM signal with a noise level of $\tau = 10$ nT, alongside the denoised results obtained from various methods, including EK-AKF, SFSDSAnet, C-GAN, TEMDnet, and TEMSGnet. Based on the observations in Fig. 10, the following conclusions can be drawn.

1) The original noisy signal exhibits significant noise interference, making it challenging to distinguish the true underlying signal.
2) The denoising result of the SFSDSA, C-GAN, and TEMDnet shows some improvement compared to the EK-AKF signal. However, there are still residual noise artifacts present.
3) In contrast, the denoising result from TEMSGnet exhibits a significant reduction in noise, effectively recovering the true signal from the noisy input.
4) This visual comparison demonstrates the superior denoising performance of TEMSGnet, as it effectively suppresses the noise and enhances signal clarity.

To further validate the generalization capabilities of VDM-based methods, especially in the context of real-world noise scenarios, we conducted additional experiments in which our deep learning models were trained on a dataset with a noise level of 10 nT and subsequently tested on signals contaminated with sferics noise at a similar noise level of 10 nT. The results of these experiments are visually represented in Fig. 11. Upon careful observation, it is evident that the traditional method EK-AKF struggles to effectively remove sferics noise. In contrast, deep learning methods, namely, SFSDSA, C-GAN, and TEMDnet, show a marked improvement in suppressing sferics noise to a certain extent. Remarkably, TEMSGnet, due to its distinctive network characteristics that prioritize generating smooth signal curves, exhibits even more effective suppression of sferics noise. We believe that these supplementary

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

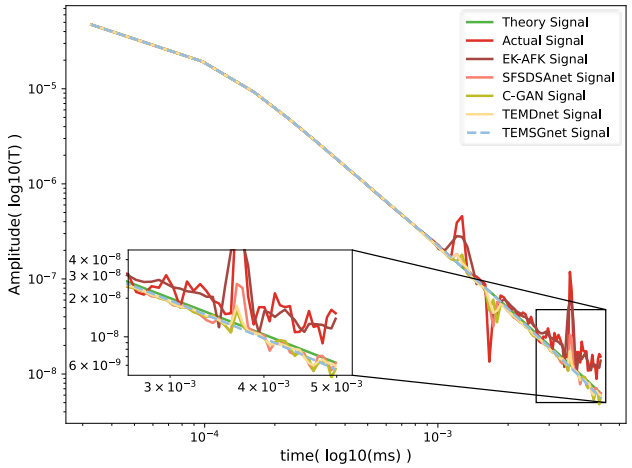


Fig. 11. Comparison of the theoretical noiseless signal and the actual signal and methods' denoising results in the logarithmic domain. The actual signal represents the noisy signal contaminated with sferics, the theory signal represents the theoretical noiseless signal, and the rest of the curves represent the denoised signals of different networks trained on the training set with a noise level of 10 nT and randomly added sferics noise.

TABLE IV

SNR RESULTS OF DIFFERENT METHODS ON SIMULATED SATEM SIGNALS AT NOISE LEVELS $\tau = 10$ NT, $\tau = 20$ NT, $\tau = 30$ NT, $\tau = 40$ NT, AND $\tau = 50$ NT. THE VALUES REPRESENT THE SNR IN DECIBELS (DB), AND THE BEST DENOISING RESULT IS HIGHLIGHTED IN BOLD. NOTE THAT ALL THE DEEP LEARNING METHODS ARE TRAINING UNDER NOISE LEVEL RANGING FROM 10 TO 50 NT

Noise Level(T)	$\tau = 10$	$\tau = 20$	$\tau = 30$	$\tau = 40$	$\tau = 50$
EK-AKF	38.37	33.83	32.41	31.19	30.41
SFSDSA	41.71	39.89	37.53	36.79	34.74
C-GAN	41.96	40.16	39.63	38.55	36.42
TEMDnet	43.45	40.12	40.53	40.13	39.88
TEMSGnet	**50.36**	**49.65**	**48.37**	**47.78**	**47.69**

experiments provide valuable evidence to support the robust generalization capabilities of VDM-based methods, particularly in real-world noise scenarios.

3) Comparison With Mixed Noise Levels on the Simulated Dataset and Other Methods: In Section IV-D2, we compared the performance of our proposed method and the baseline methods (all the deep learning methods are training under a single noise level 10 nT). To further investigate the relationship between the performance of the baseline methods and our proposed method, we introduced mixed noise levels in the training and testing simulated datasets (with noise levels ranging from $\tau = 10$ nT to $\tau = 50$ nT, the number of signal samples is consistent across noise levels) and used them as the dataset for training and testing the networks. Note that all the deep learning methods are trained and valid at the same train and valid dataset. Table IV shows the SNR scores of all methods under different noise levels. Based on the observations, we can draw the following conclusions.

1) From the experimental results, it can be observed that, except traditional method EK-AKF, the denoising performance of all deep learning methods improves when using a dataset with greater diversity.
2) Overall, TEMSGnet demonstrates competitive performance compared to SFSDSA, C-GAN, and TEMDnet. This may be attributed to the fact that our method effectively covers the encoding region of the latent space.
3) It is worth noting that TEMSGnet achieves good denoising results in both cases with higher and lower noise levels. This can be attributed to the strong constraint generation modeling capability of the TEMSGnet network on the clean signal.

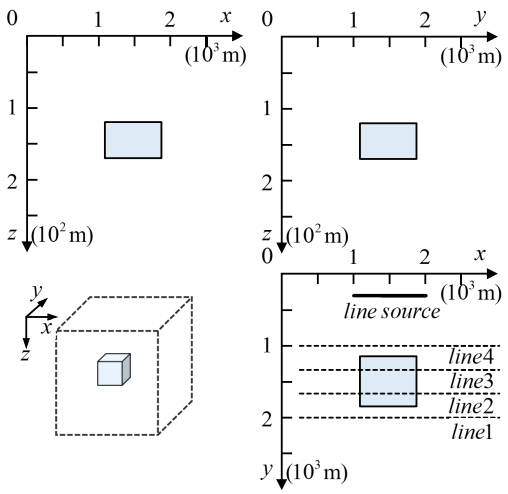


Fig. 12. Three views of the model with two layers that have a single anomaly and 3-D drawings. We choose the results of line 2 as the data for the subsequent experiments.

Finally, we conducted a comparative analysis of the time consumption of different methods on the valid dataset, as illustrated in Table V.

1) To begin with, it is readily apparent that deep learning methods, benefiting from GPU acceleration, exhibit remarkably swift inference times after training. Given the relatively straightforward network architecture of SFSDSA, its time consumption is notably less than that of C-GAN, TEMDnet, and TEMSGnet. Furthermore, owing to TEMSGnet's reliance on the VDM, both its training and inference processes necessitate iterative steps, and its network structure, being more intricate compared to TEMDnet, incurs a longer processing time.
2) The high computational costs of C-GAN are primarily due to their complex architecture, adversarial training, and the integration of conditional information. Optimizing these networks and dealing with conditional aspects contribute to increased time and resource requirements.
3) Despite the fact that our proposed TEMSGnet network leads to increased training and inference times (primarily attributable to the specific nature of VDM), it showcases outstanding generalization capabilities. This attribute proves particularly valuable when dealing with TEM signals that, although precious, have undergone limited prior processing.

4) Results for the Multipoint Denoising of the SATEM Signal on Simulated Dataset: We selected a model with two layers that have a single anomaly. The dimensions of the model are given as follows: length and width (x- and y-directions) are both 3000 m, and the depth (z-direction) is 300 m. As shown in Fig. 12, the background resistivity of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

TABLE V

TIME CONSUMPTION OF DIFFERENT METHODS FOR DENOISING WITH SATEM SIGNALS. NOTE THAT THE CENTRAL PROCESSING UNIT (CPU) IS INTEL I7 QUAD-CORE 2.9 GHZ AND THE GPU IS NVIDIA GEFORCE RTX 3080TI

Method	Training	Noise Prior	Device Detail	Training Time (in Hours)	Running Time (in Seconds per 1024 signals)
EK-AKF	-	Needless	CPU	-	282.39
SFSDSA	Need	Need	GPU	3.3	0.15
C-GAN	Need	Need	GPU	6.8	0.76
TEMDnet	Need	Need	GPU	4.2	0.54
TEMSGnet	Need	Need	GPU	7.5	0.97

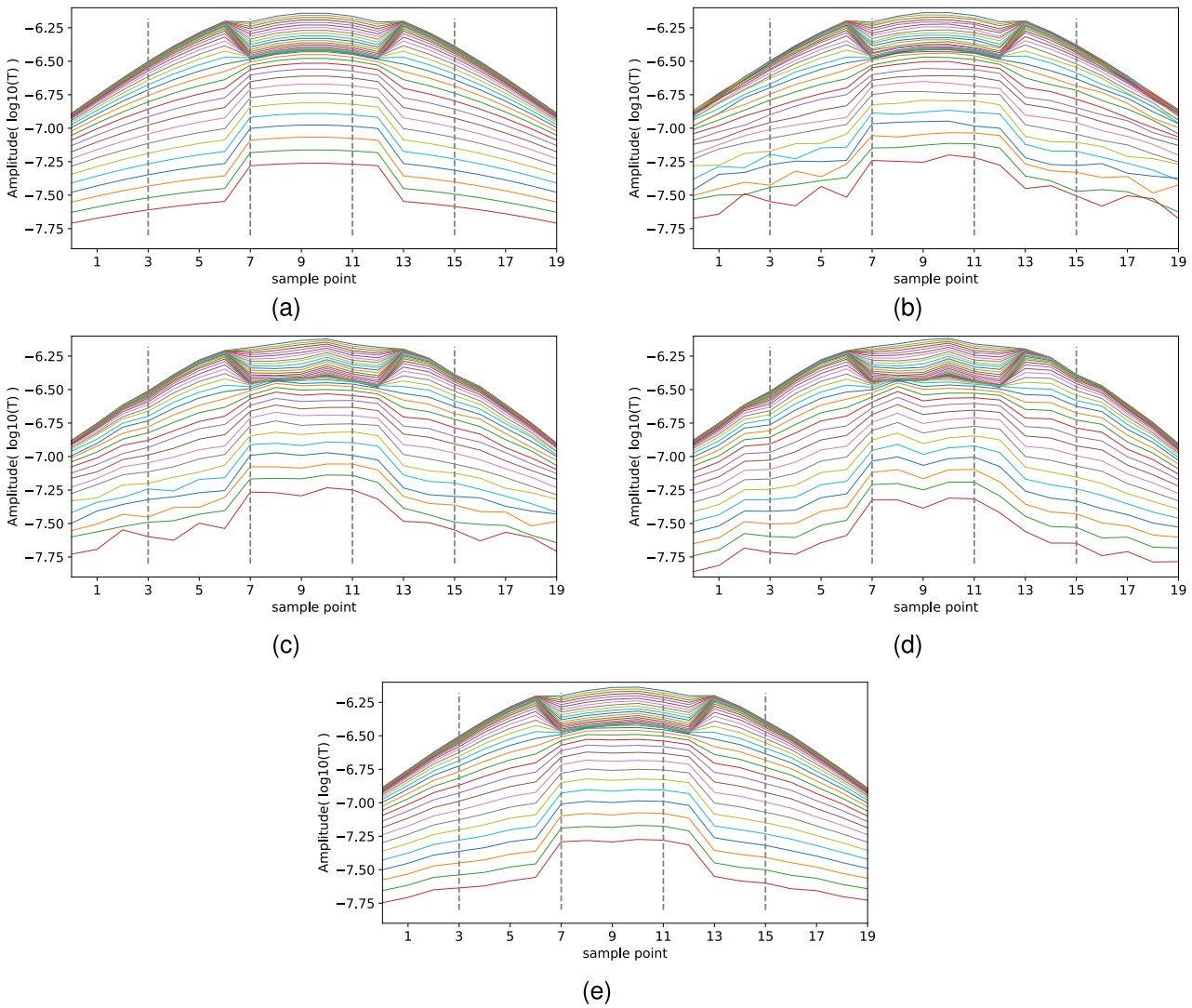


Fig. 13. Results for the multipoint denoising of the SATEM signal. (a) Theory signal. (b) Noised signal. (c) Result of C-GAN denoising. (d) Result of TEMDnet denoising. (e) Result of TEMSGnet denoising.

model is 200 Ω. The anomaly is a low resistivity anomaly in the shape of a rectangular prism, with dimensions of 800 m (length and width), 50 m (thickness), and a resistivity of 50 Ω. The anomaly is located at the center point with coordinates (1500, 1500, 150). In the forward modeling setup, the length of the current source is set to 1000 m. The center position of the line source is (1500, 100), parallel to the x-axis, with a current intensity of 20 A. The model is divided into a grid of size 100×100. The starting thickness is set to 1 m, and the thickness increases geometrically with a ratio of 1.1 for each layer. The time range is from $5.20833e^{-6}$ to 0.005 ms, with a total of 64-time windows. The lines are parallel to the line source and have a length of 3000 m. There are 20 measurement points evenly distributed along the lines.

Building upon prior experiments and the findings from earlier research [7], we observe a marked superiority in TEMDnet's performance compared to EK-AKF and SFSDSA. Therefore, in our subsequent experiments, we exclusively employ C-GAN and TEMDnet as the benchmark for comparison.

Fig. 13(a) shows the results for the multipoint denoising of the SATEM signal. Fig. 14 shows the SATEM signals from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

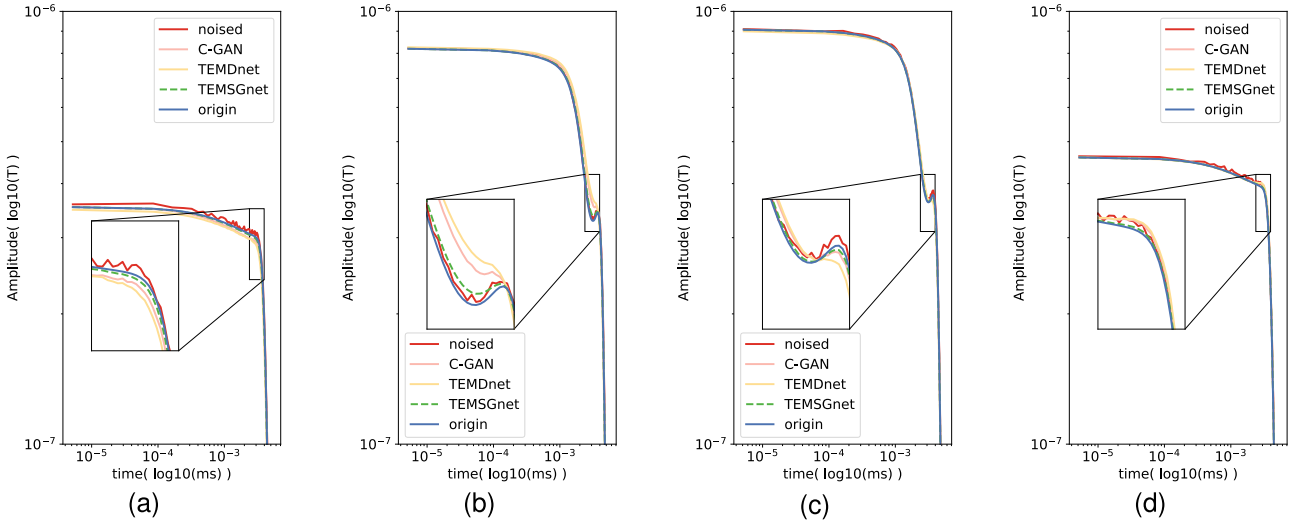


Fig. 14. Results for the single-point denoising of the SATEM signal. (a) Point 3. (b) Point 7. (c) Point 11. (d) Point 15.

the single point 3, 7, 11, and 15 corresponding to the dashed lines in Fig. 13. By observing Fig. 15(a), we can see that the forward modeling result of the theoretical model is relatively smooth in the horizontal direction, and the concave part in the middle corresponds to the location of the anomaly. When noise is added, the 2-D forward curve [Fig. 13(b)] exhibits jittering in the horizontal direction and overlapping of different curves. The denoising result using C-GAN [Fig. 13(c)] and TEMDnet [Fig. 13(d)] effectively suppresses the overlapping of curves, but there is still some fluctuation in the horizontal direction. The result using our proposed method [Fig. 13(e)] shows excellent suppression of curve overlapping and better smoothness in the horizontal direction compared to TEMDnet. In conclusion, the denoising effect of TEMSGnet is superior to C-GAN and TEMDnet in terms of suppressing curve overlapping and achieving smoother results in the horizontal direction.

To further investigate the denoising results, the inverted results of the theoretical clean curve, as well as the denoised curves obtained by C-GAN, TEMDnet, and TEMSGnet, are examined. The inversion parameters are set as follows. The 1-D inversion is performed with 30 layers, a starting thickness of 1 m, a background resistivity of 200 Ω, three iterations, and a minimum fitting error of 0.01. The inversion results are shown in Fig. 15. Fig. 15(a)–(d) shows the inversion results for the theoretical clean curve, the denoised curve using C-GAN, the denoised curve using TEMDnet, and the denoised curve using TEMSGnet, respectively. The black dashed boxes represent real anomalies. By comparing the inverted results with the theoretical model, it can be observed that both denoising methods are able to recover the general shape of the anomaly. However, TEMSGnet achieves a more accurate and detailed inversion result compared to C-GAN and TEMDnet, with a closer match to the true model. This indicates that TEMSGnet has better denoising performance and is able to preserve more information about the subsurface anomaly during the denoising process.

5) Comparison With Other Advanced Methods on Real Geological Areas: The dataset used for transfer learning encounters a challenge related to data constraints, primarily stemming from the limited availability of real SATEM signal data. We could only employ approximately 160 data points extracted from real-world area sidelines1, as illustrated in Fig. 8, for training. It is important to note that the traditional method used for preprocessing samples with noise is the wavelet transform. The training epoch for transfer learning is 50. The relatively modest dataset size poses a considerable challenge in enhancing model performance for the target task. To address this issue, we will rely on a pretrained model. This model was originally trained on a simulated dataset encompassing a substantial volume of data and a wider spectrum of samples, enabling it to capture a broader array of features and knowledge, which we can then adapt to our specific target task. For the fine-tuning of real-world datasets, we used a multistep process. Initially, we applied the wavelet transform method for data filtering. Subsequently, we manually curated the data to create a more realistic dataset, comprising approximately 160 samples. This manual curation process took approximately 20 h. For all deep learning methods, we have fine-tuned them on the actual workspace.

Since the signals from sampled points exhibit substantial noise in the late stage, we employ the wavelet transform for denoising after their superposition, as depicted in Fig. 16(a). Despite the application of the wavelet transform, the signals remain characterized by numerous crossovers in the middle and late stages, rendering them unsuitable for direct inversion. Consequently, manual adjustments are necessitated. It is imperative to note that the actual signal used for deep learning denoising has already undergone wavelet transformation. The proposed deep learning model builds upon the foundation of wavelet transform denoising, introducing further optimizations. Anticipating that our network model can notably diminish the need for manual adjustments through transfer learning, we envision an enhanced practical applicability of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

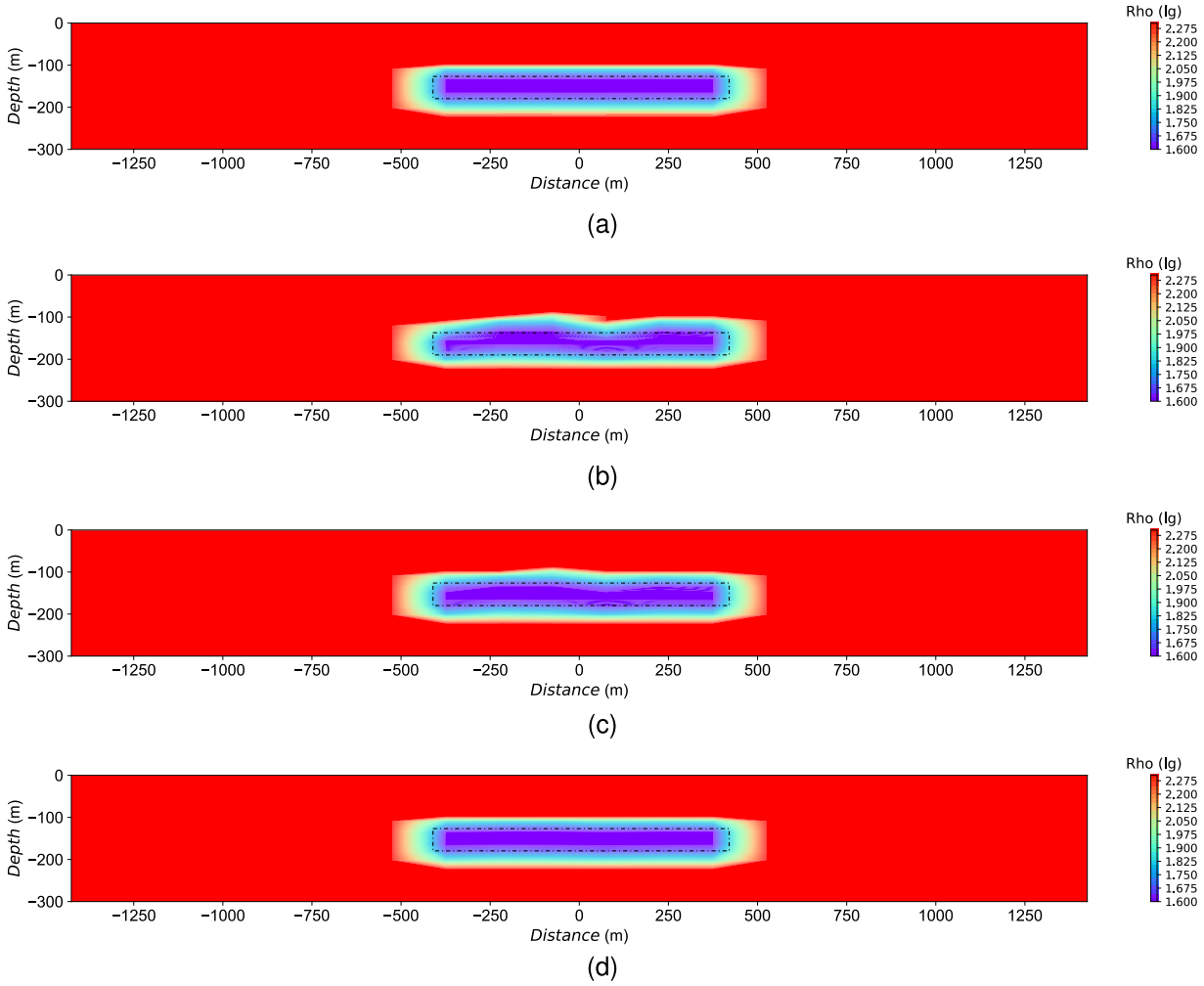


Fig. 15. Inversion results for the theoretical clean curve, the denoised curve using TEMDnet, and the denoised curve using TEMSGnet. (a) Theory. (b) C-GAN denoising. (c) TEMDnet denoising. (d) TEMSGnet denoising.

our approach. This clarification aims to elucidate the practical advantages of our method.

The sideline is equipped with 156 sampling points, with a length of 600 m. The distances from the sideline to the line source are 200 m. In the following analysis, we will compare the denoising results of different networks for the sampling points with indices 20, 60, 100, and 140.

Fig. 16 presents the acquired sideline 2 signals from the actual field area, the manually adjusted results, and the denoising results obtained using C-GAN, TEMDnet, and TEMSGnet. First, it is evident that the SATEM signals in the later stages are heavily contaminated by noise, resulting in significant cross-coupling phenomena in the actual data and hindering geological feature analysis. Most methods demonstrate limited effectiveness in denoising the shallow subsurface features, which restricts the estimation of deeper underground information. However, TEMSGnet exhibits superior denoising performance in the later stages compared to C-GAN and TEMDnet, with fewer instances of cross-coupling. Moreover, the denoised results from TEMSGnet resemble the manually adjusted results, indicating a better denoising effect. This suggests that TEMSGnet effectively reduces noise interference in the later stages of SATEM signals, leading to improved interpretability and better preservation of important geological features. The reduced cross-coupling in the denoised results of TEMSGnet enhances the accuracy of the subsurface image and facilitates more reliable geological feature analysis. It enables a more accurate estimation of deeper underground information through inversion methods. Overall, TEMSGnet demonstrates superior denoising performance in the later stages of SATEM signals compared to C-GAN andTEMDnet and enhances the interpretability of subsurface geological structures. Fig. 17 shows the SATEM signals from the single point 20, 60, 100, and 140 corresponding to the dashed lines in Fig. 16.

The inversion results are shown in Fig. 18. The inversion parameters used in the process are as follows: the number of layers is set to 30, the initial thickness is 1 m, the background resistivity is 200 Ω, the iteration count is set to 3, and the minimum fitting error is set to 0.01. By observing the inversion results, we can see that our results of the network are highly consistent with the manually adjusted results. In addition, we notice that the estimated geological layers of the C-GAN and TEMDnet are nearly identical to ours in the shallow subsurface. However, when it comes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

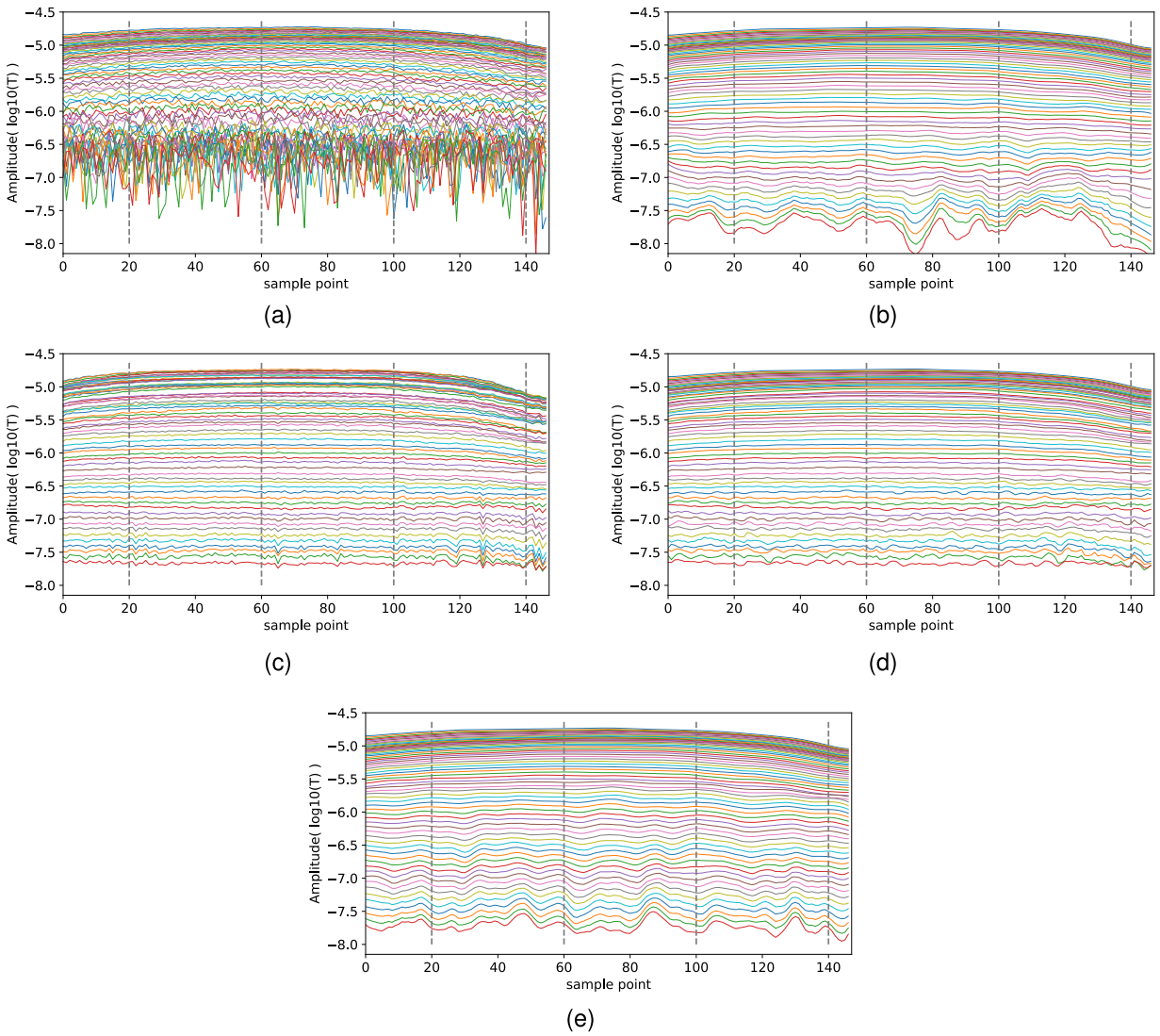


Fig. 16. Results for the multipoint denoising of the SATEM signal from the actual field area. (a) Actual signal. (b) Manual denoising. (c) Result of C-GAN denoising. (d) Result of TEMDnet denoising. (e) Result of TEMSGnet denoising.

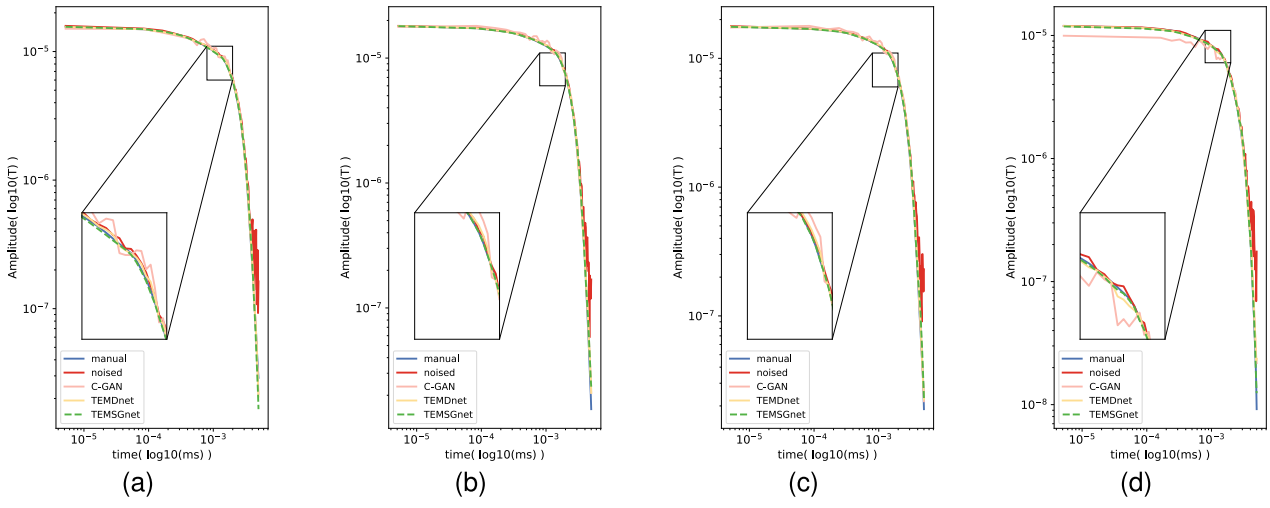


Fig. 17. Results for the single-point denoising of the SATEM signal from the actual field area. (a) Point 20. (b) Point 60. (c) Point 100. (d) Point 140.

to deeper layers closer to the bottom, the results lack the smooth variations observed in ours. In reality, geological layers tend to exhibit gradual transitions rather than abrupt changes. Hence, our results better reflect the gradual variations observed in actual geological formations, which is a more realistic representation. The similarity between the results of TEMSGnet and the manually adjusted results indicates the effectiveness of our approach in capturing the true subsurface

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

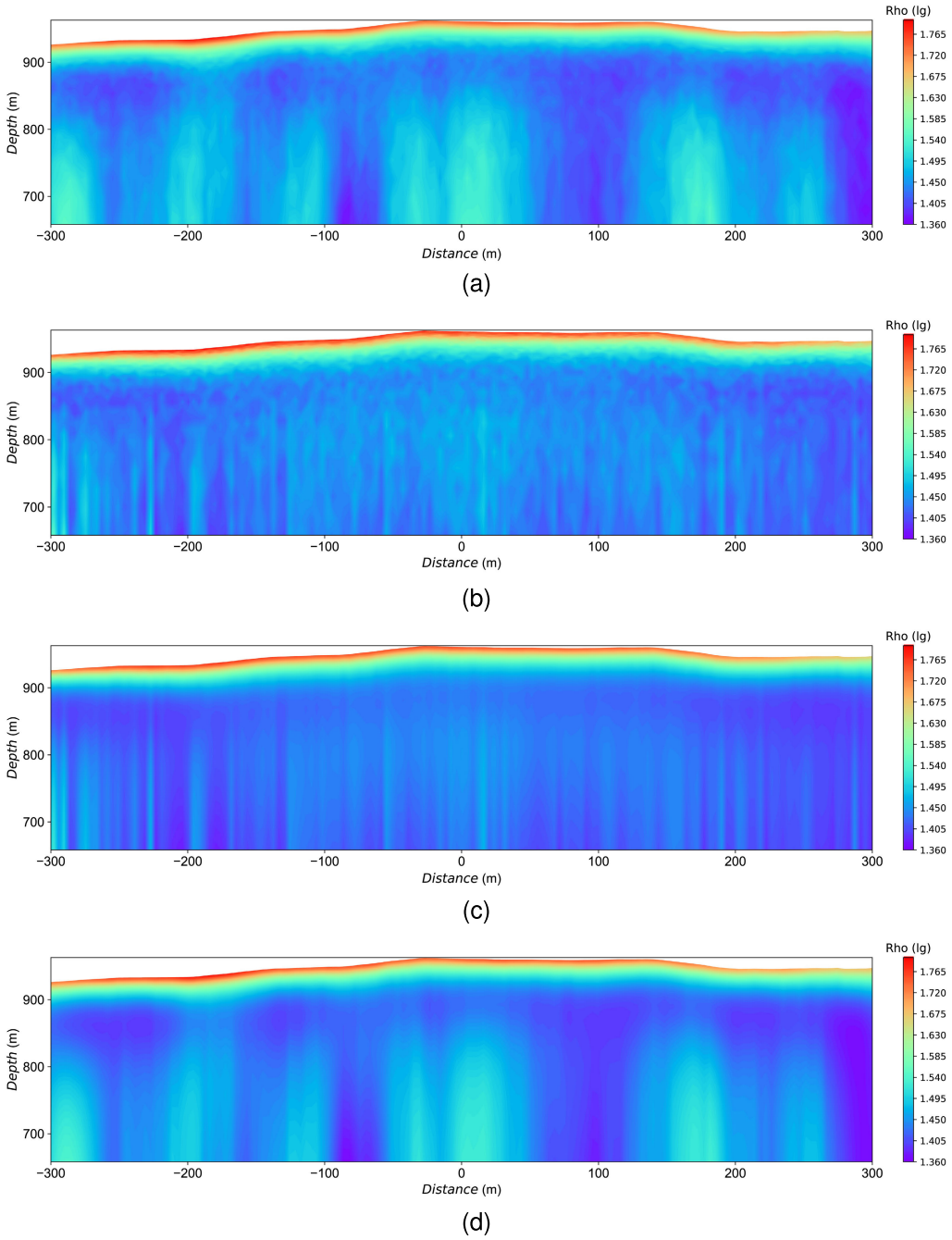


Fig. 18. Inversion results for the theoretical clean curve, the denoised curve using TEMDnet, and the denoised curve using TEMSGnet from the actual field area. (a) Manual denoising. (b) C-GAN denoising. (c) TEMDnet denoising. (d) TEMSGnet denoising.

features. This alignment further validates the accuracy and reliability of our network in capturing the geological variations. It is important to note that the quality and reliability of inversion results depend on various factors, including the quality of the input data, the chosen inversion parameters, and the complexity of the subsurface. Therefore, while our results exhibit better agreement with the true geological variations, further analysis and validation using additional geological information are essential to fully assess the accuracy and reliability of the inversion results. In conclusion, the inversion results of TEMSGnet closely resemble the manually adjusted results, and they exhibit smoother and more realistic variations compared to C-GAN and TEMDnet, particularly in the deeper regions. This demonstrates the superiority of our approach in capturing the true behavior of subsurface geological layers. In Fig. 19, we give a framework figure to explain how this method works in real-world TEM denoising.

E. Discussion

To comprehensively quantify the performance of our networks, we employ the root-mean-square error (RMSE) as an evaluation metric. All deep learning methods (SFSDSAnet, C-GAN, TEMDnet, and TEMSGnet) are trained using a simulated dataset with mixed noise level ranging from 10 to 50 nT.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

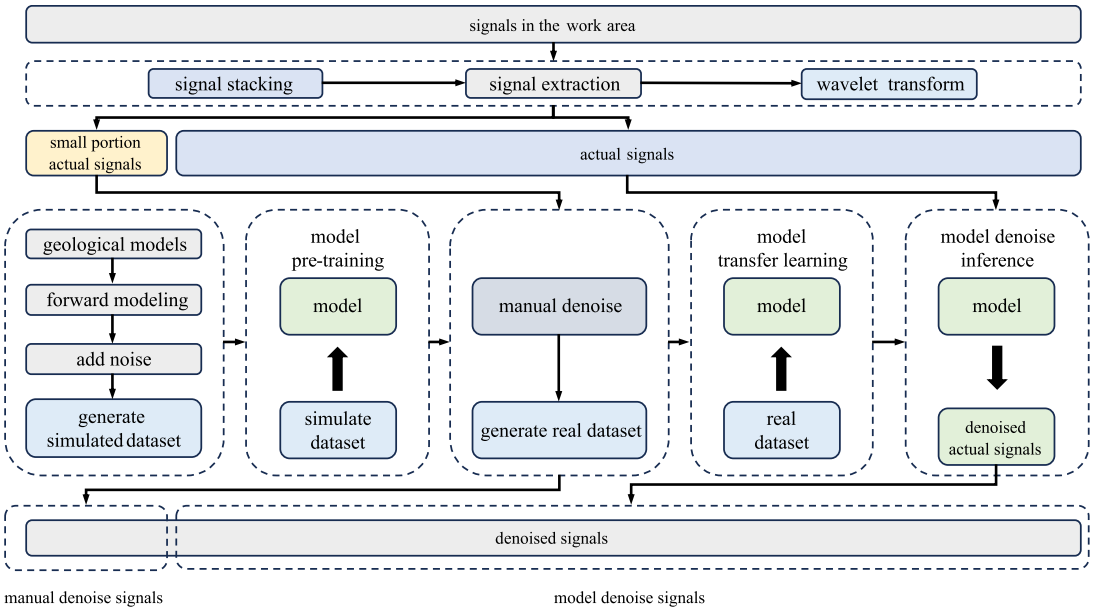


Fig. 19. Workflow for enhancing TEM signal denoising using transfer learning. The process begins with a pretrained model that has been trained on a simulated dataset, rich in data diversity and volume. Real-world SATEM signal data, approximately 160 data points, are processed using wavelet transform to reduce noise. This is followed by a manual denoising step to refine the signal further, resulting in a curated dataset of realistic samples. The pretrained model is then fine-tuned on this real dataset within an actual workspace environment, enabling the model to learn from the authentic signals. Despite initial wavelet denoising, the signals exhibit significant noise, particularly in later stages, necessitating additional manual adjustment. The deep learning model is designed to minimize these manual interventions by leveraging the inherent noise-reduction patterns learned through transfer learning, thereby improving the practical application of the model for SATEM signal analysis. In addition, considering the completeness of the process, the production process of the theoretical dataset is also given in the figure. The process includes obtaining the theoretical SATEM signal based on the forward modeling and adding noise to obtain the sample pairs.

TABLE VI

DETAILS OF SNR AND RMSE FOR EACH METHOD. SPECIFICALLY, FOR THE ORIGIN ENTRY IN THE TABLE, IT REPRESENTS THE SNR AND RMSE CORRESPONDING TO THE ORIGINAL NOISE SIGNAL. FOR THE ORIGIN-VDM ENTRY IN THE TABLE, IT REPRESENTS THE SNR AND RMSE CORRESPONDING TO THE ORIGIN-VDM

Method	Training	Constraints	SNR	RMSE
Origin	-	-	35.23	0.1350
EK-AKF	Needless	-	38.37	0.0956
SFSDSA	Need	-	41.71	0.0759
C-GAN	Need	-	43.88	0.0696
TEMDnet	Need	-	46.45	0.0603
TEMSGnet	Need	Need	50.36	0.0367
Origin-VDM	Need	Needless	8.74	2.773

For testing, each method, including EK-AKF, SFSDSAnet, C-GAN, TEMDnet, and TEMSGnet, uses a training set with noise level fixed at 10 nT. In addition, to elucidate the role of constraints, we trained an origin-VDM, which does not incorporate conditional constraint information. The training and testing procedures for this model align with the other methods. The RMSE results for each model are detailed in Table VI, revealing the following observations. For the traditional EK-AKF method, due to its limited processing capability, it achieves a partial improvement in noise suppression compared to the origin, resulting in a moderate decrease in RMSE. In contrast, deep learning methods, SFSDSAnet, C-GAN, and TEMDnet, outperform EK-AKF by leveraging their robust feature extraction capabilities, thereby demonstrating superior SNR levels and lower RMSE compared to EK-AKF. Our network, TEMSGnet, excels in denoising efficacy owing to its outstanding performance, showcasing even lower RMSE compared to the preceding deep learning methods. Furthermore, we note that the SNR level of origin-VDM is remarkably low, and its RMSE is significantly high, surpassing the SNR and RMSE of the original noise (origin). This phenomenon arises from the unconstrained nature of the origin-VDM model, generating signals with inherent randomness. This underscores the crucial necessity of constraints for TEMSGnet.

While constructing a simulated dataset can encompass a significant portion of the sample space, the unique characteristics of TEM signals prevent comprehensive coverage. Furthermore, due to the intricate nature of geological conditions, this study confirms the method's superiority in regions with accessible data. However, the challenge of sample acquisition hinders the validation of the network's denoising efficacy across diverse geographical areas. To enhance our understanding of denoising performance, it is imperative to conduct experimental validation across a broad spectrum of geological regions.

In this study, the transfer learning strategy for field-measured data may not be as flexible. However, for real-world geomagnetic data, where the signal is weak and the noise is substantial, training an excessively large model to account for all scenarios is impractical. Instead, training a base model and fine-tuning it on a small subset of lines selected from an actual work area can perform well in that specific area, albeit not entirely replacing manual labor. Nonetheless, it significantly reduces the need for manual intervention. Furthermore, the simulated data in this study are generated through forward modeling of geological formations. Although forward

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.

modeling can approximate SATEM signals to real conditions, the introduced Gaussian noise may differ from the actual signal noise. Hence, considering a more realistic noise generation method is crucial for enhancing the network's robustness. For instance, utilizing generative networks to generate noise that better matches the characteristics of real signals.

V. Conclusion

Due to the rapid decay of the early amplitude and the slower and smaller decay of the middle to the late amplitude of SATEM signals, they are more susceptible to noise interference. The early signals correspond to shallow subsurface information, while the middle-to-late signals correspond to deeper subsurface information. Therefore, the accuracy of denoising the middle-to-late signals is crucial. Traditional methods struggle to remove noise from the middle-to-late signals and often require manual postprocessing, which can be time-consuming for large datasets. Moreover, discriminative models have limited generalization ability when trained on a small number of samples, resulting in suboptimal denoising performance.

This study introduces a novel constrained VDM method specifically tailored for SATEM signals. This method guides the generation of denoised signals and effectively improves the generalization of the network to unseen data using existing training samples. Experimental comparisons confirm that our approach significantly improves the generalization ability of the network, particularly when dealing with limited training datasets, resulting in notable noise suppression for SATEM signals. Our findings offer a fresh perspective on SATEM signal denoising while effectively addressing the limitations of traditional methods and discriminative models in terms of generalization and denoising effectiveness. In summary, our proposed constrained VDM method and TEMSGnet provide a new approach to denoise SATEM signals. This leads to better noise suppression in SATEM signals and opens up new possibilities for denoising methods in the field of SATEM signal processing.

References

[1] X. Wu, G. Xue, Y. Zhao, P. Lv, Z. Zhou, and J. Shi, "A deep learning estimation of the Earth resistivity model for the airborne transient electromagnetic observation," *J. Geophys. Res., Solid Earth*, vol. 127, no. 3, Mar. 2022, Art. no. e2021JB023185.

[2] A. P. Annan and R. Lookwood, "An application of airborne geotem in Australian conditions," *Explor. Geophys.*, vol. 22, no. 1, pp. 5–12, Mar. 1991.

[3] Y. Ji, D. Li, M. Yu, Y. Wang, Q. Wu, and J. Lin, "A de-noising algorithm based on wavelet threshold-exponential adaptive window width-fitting for ground electrical source airborne transient electromagnetic signal," *J. Appl. Geophys.*, vol. 128, pp. 1–7, May 2016.

[4] Y. Ji et al., "Noise reduction of grounded electrical source airborne transient electromagnetic data using an exponential fitting-adaptive Kalman filter," *Explor. Geophys.*, vol. 49, no. 3, pp. 243–252, Jun. 2018.

[5] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

[6] F. Lin, K. Chen, X. Wang, H. Cao, D. Chen, and F. Chen, "Denoising stacked autoencoders for transient electromagnetic signal denoising," *Nonlinear Processes Geophys.*, vol. 26, no. 1, pp. 13–23, Mar. 2019.

[7] K. Chen, X. Pu, Y. Ren, H. Qiu, F. Lin, and S. Zhang, "TEMDnet: A novel deep denoising network for transient electromagnetic signal with signal-to-image transformation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5900318.

[8] D. Kingma, T. Salimans, B. Poole, and J. Ho, "Variational diffusion models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 21696–21707.

[9] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 6840–6851.

[10] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.

[11] S. Gu et al., "Vector quantized diffusion model for text-to-image synthesis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10696–10706.

[12] W. Wang, Y. Huang, Y. Wang, and L. Wang, "Generalized autoencoder: A neural network framework for dimensionality reduction," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2014, pp. 496–503.

[13] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.

[14] D. P. Kingma, T. Salimans, R. Jozefowicz, X. Chen, I. Sutskever, and M. Welling, "Improved variational inference with inverse autoregressive flow," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 1–9.

[15] X. Yang, "Understanding the variational lower bound," *Variational Lower Bound, ELBO, Hard Attention*, vol. 22, pp. 1–4, Sep. 2017.

[16] R. Ranganath, D. Tran, and D. Blei, "Hierarchical variational models," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 324–333.

[17] A. Klushyn, N. Chen, R. Kurle, B. Cseke, and P. van der Smagt, "Learning hierarchical priors in VAEs," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–10.

[18] C. K. Sønderby, T. Raiko, L. Maaløe, S. K. Sønderby, and O. Winther, "Ladder variational autoencoders," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 1–9.

[19] J. R. Norris, *Markov Chains*, vol. 2. Cambridge, U.K.: Cambridge Univ. Press, 1998.

[20] F. Csikor, B. Meszena, B. Szabo, and G. Orban, "Top-down effects in an early visual cortex inspired hierarchical variational autoencoder," in *Proc. SVRHM NIPS*, 2022, pp. 1–15.

[21] D. J. Rezende, S. Mohamed, and D. Wierstra, "Stochastic backpropagation and approximate inference in deep generative models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2014, pp. 1278–1286.

[22] C. Luo, "Understanding diffusion models: A unified perspective," 2022, *arXiv:2208.11970*.

[23] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[24] Y. Wu and K. He, "Group normalization," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[25] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.

[26] A. Sherstinsky, "Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network," *Phys. D, Nonlinear Phenomena*, vol. 404, Mar. 2020, Art. no. 132306.

[27] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[28] S. Wang, B. Z. Li, M. Khabsa, H. Fang, and H. Ma, "Linformer: Self-attention with linear complexity," 2006, *arXiv:2006.04768*.

[29] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[30] M. Mirza and S. Osindero, "Conditional generative adversarial nets," 2014, *arXiv:1411.1784*.



Fei Deng received the Ph.D. degree in Earth exploration and information technology from the College of Information Engineering, Chengdu University of Technology, Chengdu, China, in 2007.

Since 2004, he has been with the College of Computer and Network Security, Chengdu University of Technology, where he is currently a Professor. His research interests include artificial intelligence, deep learning, and computer graphics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.



Bin Wang received the B.Sc. degree from the College of Computer Science and Cyber Security, Chengdu University of Technology (CDUT), Chengdu, China, in 2022, where he is currently pursuing the M.Sc. degree in computer science and technology.

His research interests include intelligent geophysical data processing, computer vision, and applications of deep learning.



Peifan Jiang (Graduate Student Member, IEEE) received the Master of Electronic Information degree in computer technology from Chengdu University of Technology, Chengdu, China, in 2023, where he is currently pursuing the Ph.D. degree in Earth exploration and information technology.

His research interests include applications of deep learning, intelligent geophysical data processing, and electromagnetic data forward modeling and inversion.



Xuben Wang received the Ph.D. degree in geophysics from the College of Geophysics, Chengdu University of Technology, Chengdu, China, in 2002.

He is currently a Senior Professor with the College of Geophysics and the Director of the Key Laboratory of Earth Exploration and Information Technology of Ministry of Education, Chengdu University of Technology; and the Executive Director of Chinese Geophysical Society, Beijing, China. He is also the Director of the Innovation Consortium for Geophysics and Continental Dynamics, Beojing, and a Senior Visiting Scholar of the Commonwealth Scientific and Industrial Research Organization (CSIRO), Canberra, ACT, Australia. His main research interests include magnetotellurics and airborne electromagnetic forward modeling and inversion theory, and multiphysics interpretation of continental dynamics.



Ming Guo received the B.S. degree in geophysics and the M.Sc. degree in Earth exploration and information technology from Chengdu University of Technology, Chengdu, China, in 2017 and 2021, respectively, where he is currently pursuing the Ph.D. degree.

His research interests include data processing and inversion methods of electromagnetic geophysics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:41:06 UTC from IEEE Xplore. Restrictions apply.